

Meeting the challenges of innovation in management education: the case for interactive simulations

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ABSTRACT

This article responds to recent calls for innovation in management education, one objective of which is for business schools to enhance their graduates' capabilities for innovation on the job. We describe an innovative computer-based simulation lab integrated into a business curriculum as an example of a powerful learning environment; and show how interactive simulations integrate core findings from research on learning to improve the quality and kinds of learning that can be achieved. We argue that simulations can make an important contribution to the development of the new skills demanded of contemporary management graduates.

BACKGROUND

Our article responds to recent calls for innovation in management education to prepare students to manage in rapidly changing, globally competitive environments. *The Lisbon Strategy – Making Change Happen* (Commission of European Communities, 2002: 7) highlighted that:

Research, innovation and education are at the heart of the knowledge-based economy.

Europe's ability to produce, diffuse and use knowledge effectively relies heavily on its capacity to carry out quality research, to produce highly educated people and for its firms to be engaged in a continuing process of innovation.

The perceived importance of these activities is often attributed to the increased reliance on human capital for advancements such as generating productivity gains or competitive advantage. For example, the Association for the Advancement of Collegiate Schools of Business (AACSB) emphasized that "business schools play a pivotal role [in society] by developing effective leaders and providing support for the engine driving sustainable growth..." (AACSB Task Force, 2010: 3).

The AACSB Report also emphasizes the importance of preparing students to lead and manage innovation from concept to implementation. The challenges associated with novel products and processes will require innovative managers. Correspondingly, management education and training must be open to new methods which emphasize learning how to learn. This will enhance graduates' ability to learn on the job and foster innovation.

In contrast to management and business success stories of the past (which were often based on the impact of a particular product or process innovation), current thinking emphasizes the need for on-going innovation, a primary source of dynamic comparative advantage. One widely-used and successful approach to fostering continuous innovation is the use of evidence-based practice. The systematic use of data and research to guide innovations and improvements in safety and efficiency in health care is well-known, and has been achieved through 'best practice' guidelines based on rigorous reviews of the research literature, and by prospective data collection and analysis. Rousseau and McCarthy (2007) have argued on behalf of building a culture of evidence-based practice in management, and that management education must also be evidence-based.

Their argument is important for several reasons. First, expectations for evidence-informed teaching are rarely articulated. Instead expectations about the conduct of teaching are usually defined by administrative and student opinions (most often through the questions asked in teaching evaluations). Second, there is robust evidence that what students learn, and how well they can subsequently use this knowledge, is influenced significantly by conditions such as teaching methods and course design. Thirdly, many students perceive a significant gap between academic and real world practices arising from differences between the substance and methods of conventional academic teaching and learning and the demands of real world practice.

Closing this gap to ensure that students can apply research to real world problems is a perennial challenge for professional education. One of the reasons for the gap is that conventional teaching methods can have unintended consequences. For example, academic courses are carefully

structured, the presentation of problems is simplified to highlight the issues being studied, problems are linked to appropriate resources and solution strategies, and performance is monitored and assessed by an instructor. As a consequence, students are not always ready for the problems that arise in practice which can present quite differently. Professionals are rarely provided with: neatly defined problems; identification and access to relevant information; or cues about the appropriate problem-solving technique to use.

Meeting the Challenges for Management Education

Given the importance of innovation, how can management education best instill capabilities for innovation in students? Innovations are inherently unpredictable and likely to produce novel situations and problems. Management education must prepare students to apply their existing management knowledge and skills creatively to solve new problems, evaluate alternatives and think critically; as well as motivate and lead others to implement innovative processes and products. Research on student learning suggests that certain types of instructional practice will be especially useful for inculcating these capabilities. The statement of general research-informed educational principles from the global accreditation body for business schools (AACSB, 2012: 57) notes that:

The most effective learning takes place when students are involved in their educational experiences. Passive learning is ineffective and of short duration. Faculty members should develop techniques and styles that engage students and make students responsible for meeting learning goals. Many pedagogical approaches are suitable for challenging students in this way – problem-based learning, projects, simulations, etc. Faculty members should find such approaches that are suited to their subject matter, and should adopt active learning methodologies.

The report also articulates expectations that learning should be active, collaborative and cooperative, and informed by frequent and timely feedback. Beyond this, the Standards encourage instructional innovations to take advantage of “new technology-mediated and action-learning

practices” (p.55), and emphasize the value of “practice field situations where students can take risks and then learn from their successes and failures” (p.57).

The expectations described above are necessarily abstract to address teaching across the broad array of disciplines which contribute to management education. However, the abstraction can be a barrier to implementation by instructors who follow a ‘default’ model of instruction (often based on educational practices they experienced as students) which may not well-suited to current needs. For example, Rousseau and McCarthy (2007) suggest that “current use of management cases as a teaching tool tends to deemphasize instruction in evidence-based principles”. The challenge for business education is therefore two-fold: first, to increase adoption of appropriate evidence-informed educational approaches that produce graduates with the capabilities that underlie successful innovation; second, the methods themselves must be efficient for instructors to learn and use, and compelling for students.

In the following section, we argue that interactive simulations can make a strong contribution to contemporary management education. As an example we describe a lab which supports efficient integration of interactive simulations into the curricula of several business education programs. In the third section, we summarize how well-planned interactive simulations can advance the goal of educating innovative management graduates. In the final section we provide some concluding comments.

INTERACTIVE SIMULATIONS AS POWERFUL LEARNING ENVIRONMENTS

To bridge the ‘theory-practice gap’ and to manage innovation, students need to go beyond learning to understand material conceptually; they need to learn *how* to use their knowledge and skills in complex, real-time environments; and to develop skills in finding appropriate data and tools. Since the environment and management problems associated with innovations are dynamic and often novel, students need to learn ‘how to learn’, and how to learn from experience. This requires learning to monitor one’s own behavior, track progress and make plans, that is, to direct and control

one's own learning. De Jong and van Joolingen (1998) have classified these as regulatory processes.

In many professional education programs, these capabilities are developed through supervised experience in real-world settings. In industry, tools such as flight simulators are sometimes employed. However, there are few clinics or teaching practice settings to enable management students to obtain such experience. Hence, the value of a facility that enables students to apply and develop what they have learned in the classroom in order to solve problems they would find in the workplace, but in a 'safe' and controlled environment.

Management education has a tradition of using classroom-based strategies to approximate aspects of practical experience. Case studies, games, role playing, access to real-time information and data, competitions and internships are more prevalent in management education than in arts and science education, for example. Some of these methods are less well suited to management disciplines which require high levels of technical skills and in which current practice is embedded in technology, such as finance and accounting. However, these challenges can be turned to educational advantage using interactive simulations. In the following sections we will illustrate how a lab has enabled the integration of simulations into a business school curriculum and explain why interactive simulations can make a special contribution to fulfilling the challenges of contemporary management education.

The Case for Interactive Simulations

The AACSB Task Report (2010) referred to the need for business education to redress a perceived overemphasis on case study methods and to adopt pedagogical methods better suited to help students develop the skills to apply what they have learned when they enter the workplace.

Similarly, Rousseau and McCarthy (2007) advocate the necessity of *practice* for developing procedural and tacit knowledge, specific competencies (such as complex problem solving), and integrating skills across different domains, and, for teaching students how to learn from experience.

Interactive simulations that require students to apply knowledge in order to achieve specific management objectives are especially useful in this respect. It is important that the simulations not be passive, that is, they should not just involve watching a simulation with no opportunity to interact both with the simulation itself and with other participants. As illustrated below for the RIT example, students' decisions benefit from interacting in both directions with the simulation, that is, real-time datalinks (RTD) from the market to their Excel models and, if appropriate, an application program interface (API) to allow decisions to be automatically sent back into the market. This interaction provides immediate feedback on the outcomes of their decisions and facilitates learning from their own level. In addition, individual student's decisions interact with all of the other participants so they learn from the actions of others and also experience the behavioral aspects of decision making in an uncertain environment.

Simulations can also help to prepare students for work after graduation more effectively. First, they extend students' learning about theory beyond 'textbook' problems by providing experience of ill-defined problems in uncertain, dynamic environments similar to those on the job. Second, simulations can be run over a period of time and thereby require students to integrate skills appropriately across the variety of tasks associated with managing complex problems over time -- such as analyzing complex situations and alternatives, finding information, problem solving, and applying or adapting appropriate methods and tools.

Early reports of computer-based simulations in management education described simulations developed for a single purpose. For example, Berg, Dickaut, Hughes, McCabe and Rayburn (1995) described a simulation of an asset-market to demonstrate how prices are influenced by accounting information. Berg *et al.* articulated several advantages of a computer-based simulation, beyond the efficiency of student participation and reporting on students' activity, which are inherent to a technological platform. Other advantages included: the resemblance between the simulation and actual institutions and assets (realism); ease of administration and participation

(simplicity); and the ability to engage students by capturing their attention and providing prompt feedback on their actions (engagement). As with much early educational software, the development costs and learning curve required to implement one-off simulations discouraged widespread uptake.

More recent advances with technology have included labs with real-time data feeds, access to historical databases, and practice with the use of industry-standard tools (for example, Alexander *et al.*, 2001; Holowczak, 2005, 2007; Coe, 2007; Coe, Kilic & Isik, 2007; Siam, 2005). These allow students to practice using information and applications from industry. This alone is not sufficient to help students learn how to apply the theory they are learning in the classroom or to narrow the gap between research, classroom theory and practice. One reason is that the industry resources and applications have been designed for professional users, rather than as educational applications that have been custom-designed to achieve particular learning objectives.

Examples of simulations custom-designed for learning objectives

The impact of simulations on university-level learning has been investigated most thoroughly in medical and health professional education. Gates, Parr and Hughes (2012) compared nursing students' performance on test items about clinical care topics that had been learned with or without a simulation, and reported that simulated topics were associated with improved test scores. A recent synthesis of research on the effectiveness of simulations in medical education reported that simulation was an effective method for teaching skills in several domains (for example, procedural, cognitive, group skills). Of even greater importance, the synthesis indicated that skills learned through simulation methods were subsequently transferred to clinical practice (McGahie, Issenberg, Petrusa and Scalese, 2010).

The advent of labs with computer-based simulations which have been designed specifically for management education is a significant advance. These labs -- which are conceptually similar to the simulation centers used in medical education -- enable simulations to be used for a variety of educational purposes throughout the curriculum, similar to the way that flight simulators are used to

train pilots. Simulations that have been designed for educational purposes can also be designed to embed many of the conditions associated with improvements in students' learning (discussed further below). An early example of a simulation lab for management education is the Rotman *Financial Research and Trading Lab* (hereafter referred to as FRTL).

The FRTL, which opened in 1999, is particularly known for its custom-designed suite of instructional products which use interactive simulations to integrate theoretical models learned in the classroom with learning cases using real-time information from simulated and actual markets. Three FRTL platforms and their associated learning cases enable simulations of increasingly complex scenarios which can be integrated into different levels of the curriculum and used flexibly for individual students, courses and extracurricular activities.¹ Some of these products are being used at universities around the world.² One advantage of the FRTL products is that the consistent interface allows students to focus attention on the academic purposes of the simulation without the added cognitive load required to learn different procedures for different kinds of simulation exercises (Sweller, 1994).

The *Rotman Interactive Trader* (RIT), <http://rit.rotman.utoronto.ca>, combines an order driven-market platform with associated learning cases which simulate risks and opportunities for various financial securities or investment and risk management strategies. The interactive capability of RIT provides rich feedback on the consequences associated with different strategies and so helps students learn to develop effective strategies to achieve the objectives of each case. Scenarios are run iteratively, and can either be randomly generated from a previously defined distribution of potential outcomes, drawn from historical outcomes, or prescribed depending on the learning goal.

For example, if instructors wish to illustrate the effects of a stock market crash, they can do so by loading and running that scenario. Students acquire experience about how markets and other

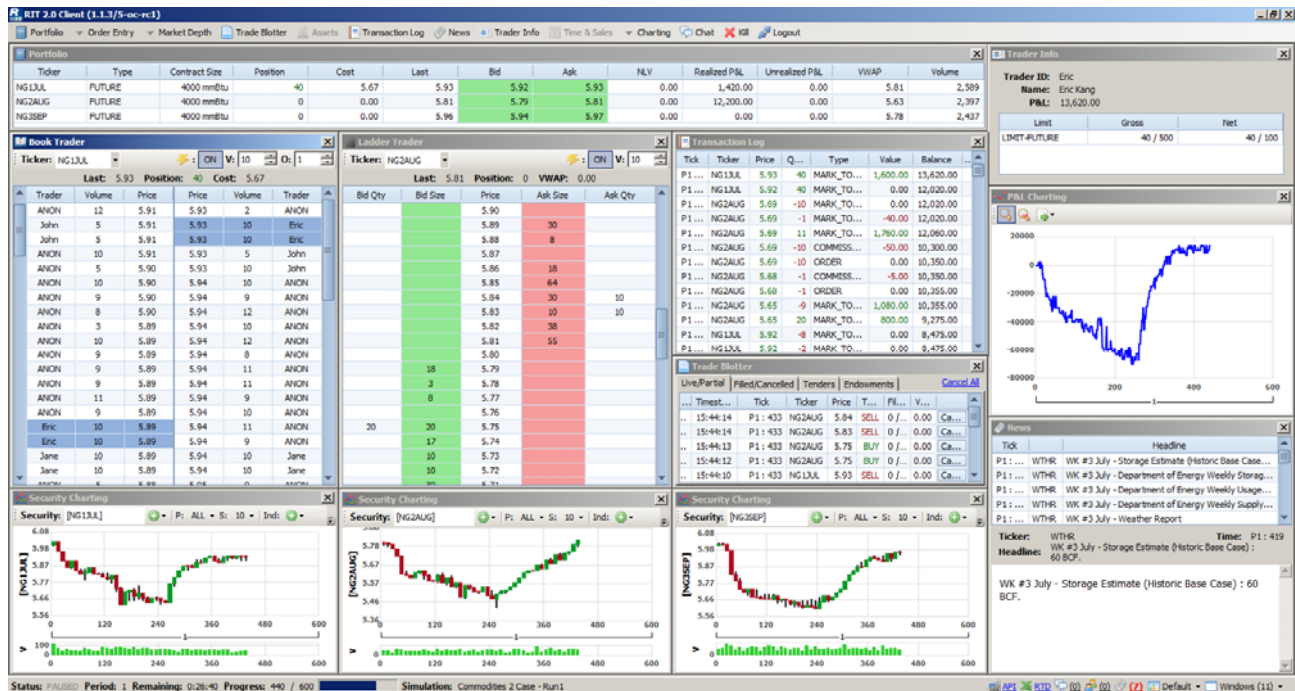
¹ An example is the international competition (RITC) which hosts competitor teams from around the world <http://ritc.rotman.utoronto.ca>

² The 43 universities currently using the RIT package include Berkeley, Duke, MIT and Stanford.

participants react to a stock market crash; they can study, evaluate, and experiment with different strategies for dealing with the situation. As they gain experience with different scenarios, students learn how to make effective, real-time decisions in a complex environment for which there is considerable uncertainty.

The RIT platform is a pure order-driven market simulator which allows the instructor to turn on programmed liquidity traders (either as informed or noise traders) depending on the objective -- for example, an instructor can choose whether or not to give students experience of the effects of an illiquid market and/or whether to introduce some mispricing in the market. Figure 1 shows a screen shot of one set of modules for the platform which can be custom designed by case and/or by user.

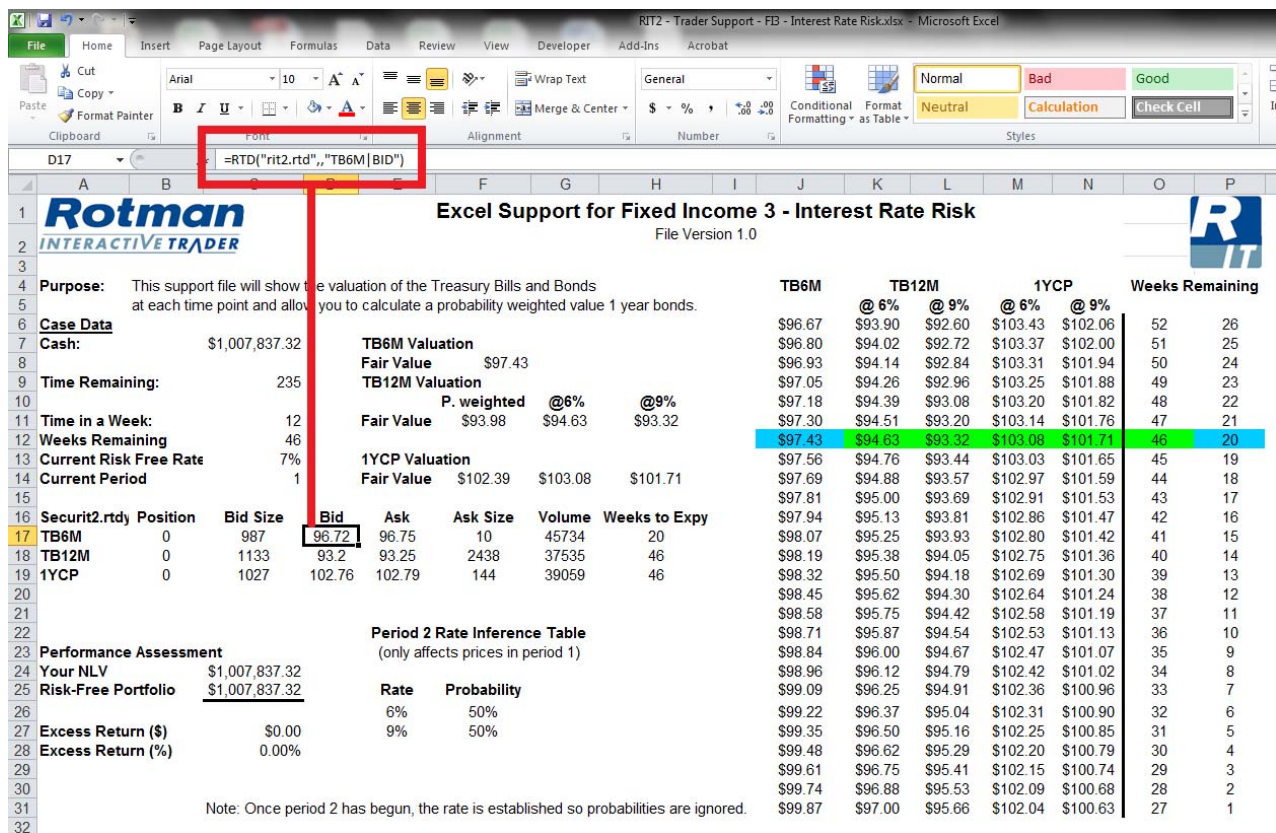
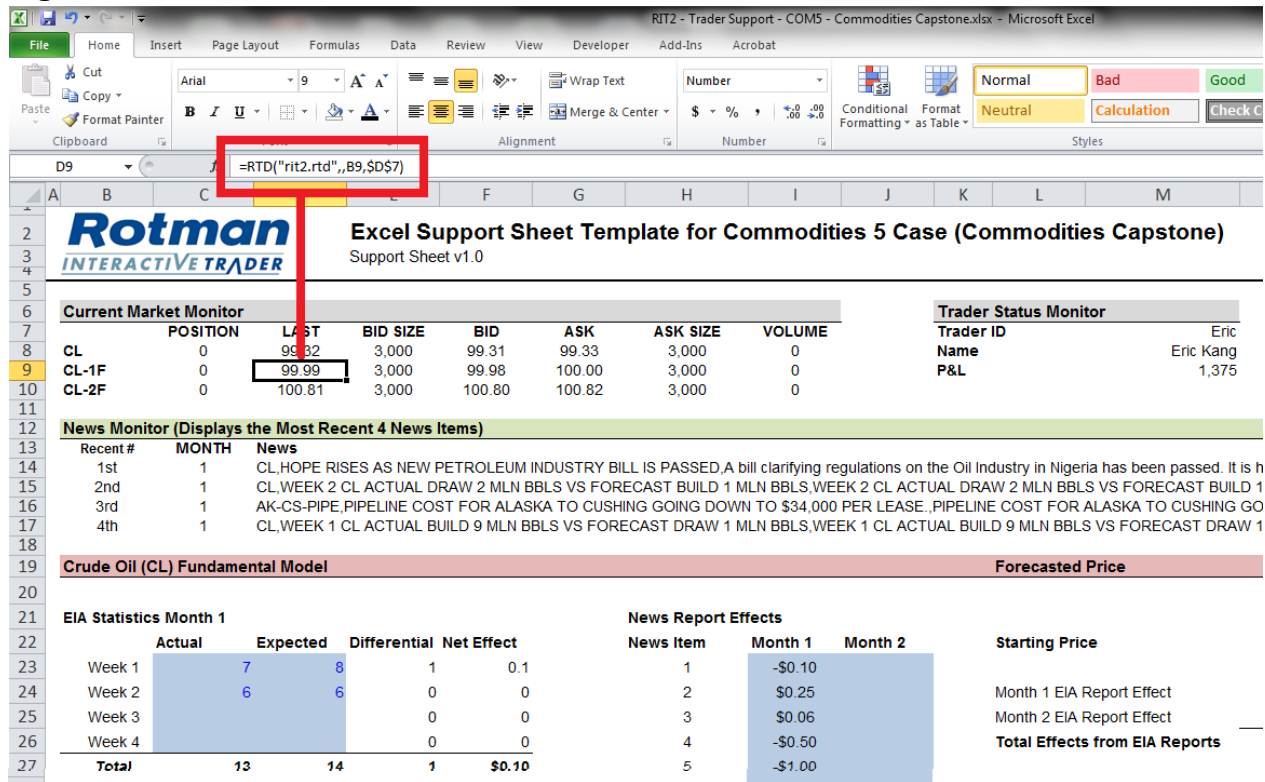
Figure 1: An example layout for the RIT platform



The RIT cases sequence from introductory (generally one source of risk) to richer cases for which the decision maker has to manage several risks. To further integrate classroom material with practice, the RIT cases have custom-designed Excel models to illustrate how theory and research evidence can be used to derive effective strategies. To facilitate this integration and interaction, Excel is connected to the RIT market simulator using real-time data (RTD) links. Figure 2 provides

screen shots illustrating example RTD links for a commodities RIT case as well as a fixed income

Figure 2: Real-Time-Data links from Excel to the RIT market simulator:



RIT case. Students determine a strategy using the theory implemented in the Excel model linked to the market and program the appropriate decision signal.

For example, in the commodities case, the Excel model calculates arbitrage opportunities across location and product type based on efficient use of assets such as pipelines, refineries, ships, etc. In a more advanced commodities case, individuals on a student team play different roles, for example, producer, refiner, trader, and must communicate with each other concerning the best decisions based on expected outcomes. In the fixed income case example (the 2nd panel of Figure 2) students use Excel to compute the fair price for TBills and government coupon bonds given interest rate risk and use the RTD links to the market to identify mispriced assets. Once students use these startup Excel models to learn how to integrate theory and practice (decision making under uncertainty), they soon learn how to extend and adapt their decision making model or even develop their own innovative model.

For some learning objectives it is useful to program a decision-making algorithm in order to clarify the logic of the decision. To facilitate such cases, the RIT utilizes an application program interface (API) so that VBA programs can direct decisions automatically back into the market. Figure 3 illustrates some screen shots showing how an arbitrageur can identify mispriced securities across alternative trading venues using a simple VBA script.

Excel models applying the relevant theory are used in other RIT cases across a wide variety of learning objectives (see <http://rit.rotman.utoronto.ca/cases.asp>). For example, they can be used to learn how to implement a portfolio insurance strategy for risk management (Figure 4), or to price a security based on incomplete and asymmetric information in order to identify securities that are over or undervalued in the market, or to determine an investment strategy balancing risk and potential return given a target retirement fund over a 20-year horizon. Students get to see, first hand, how to apply and develop the theory that they have learned in their classroom, and gain an appreciation of the value of theory and research evidence for guiding practice.

Figure 3: API Links from Excel into the RIT Market for algorithms:

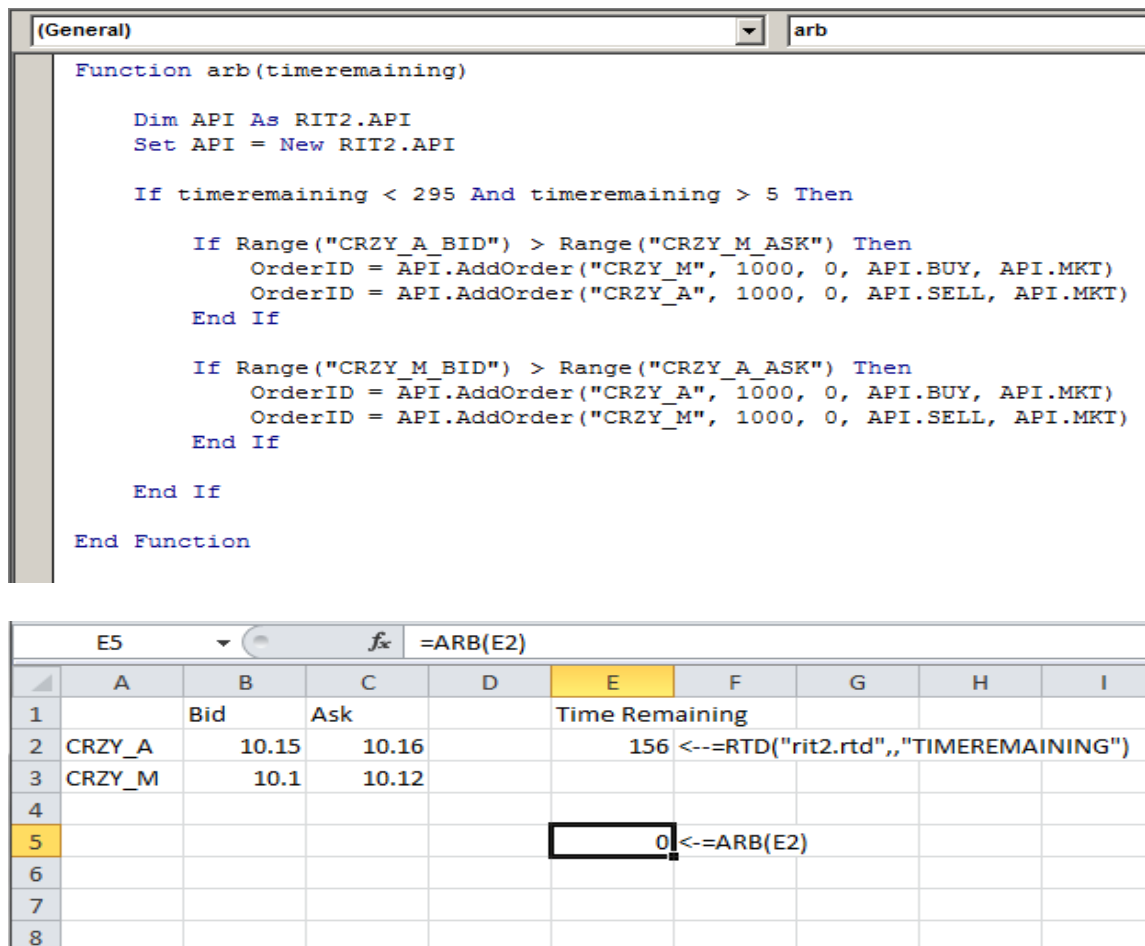
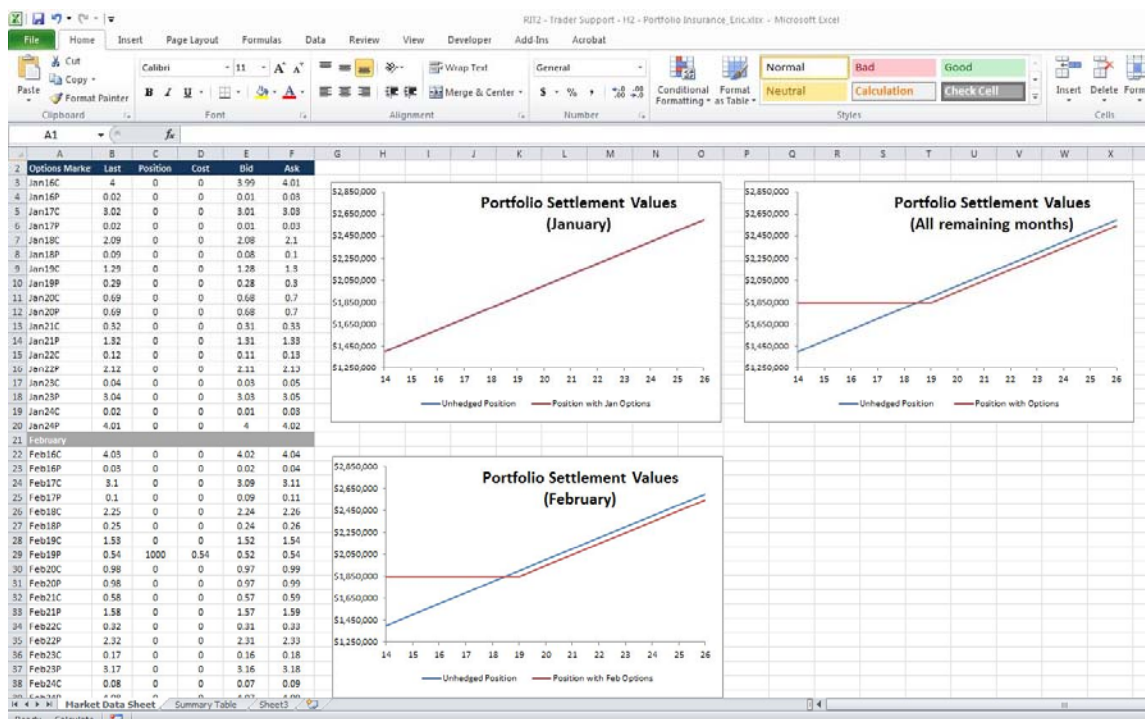


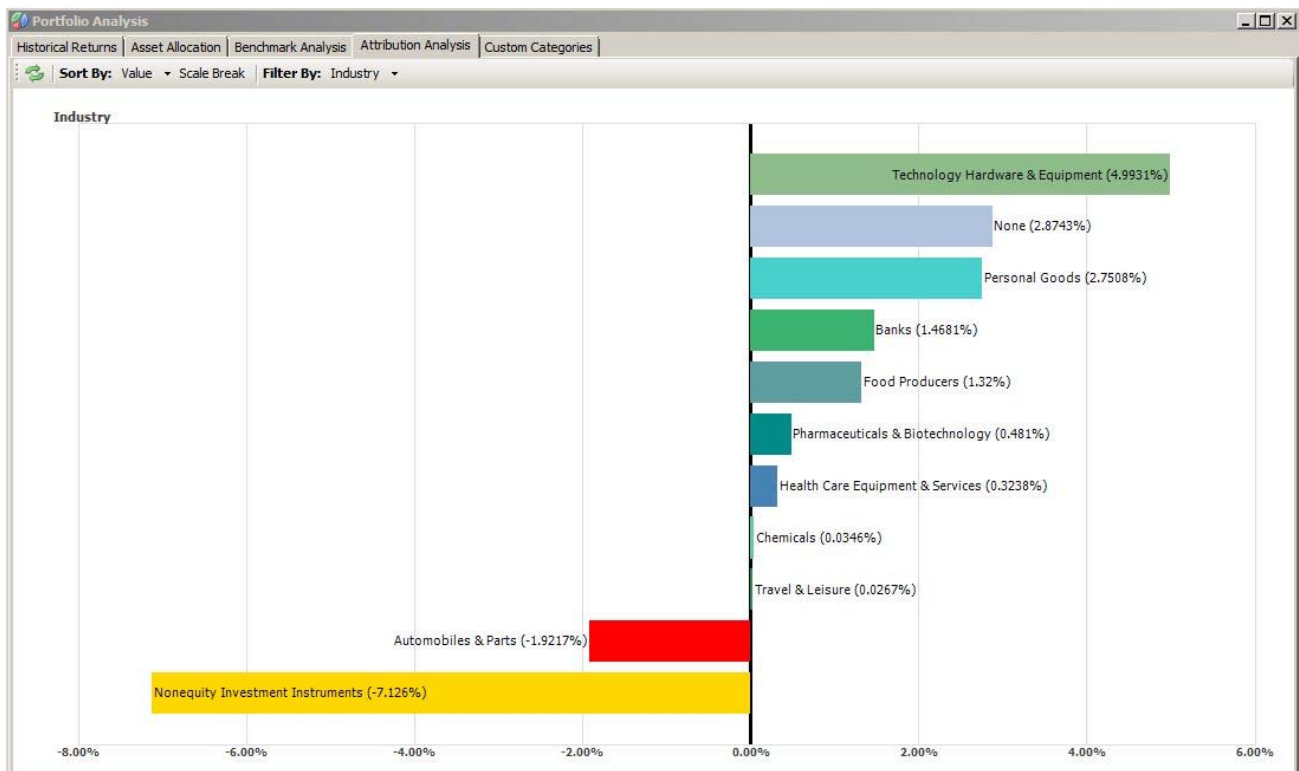
Figure 4: Example illustration of potential outcomes of a portfolio insurance decision:



The *Rotman Portfolio Manager* (RPM), <http://rpm.rotman.utoronto.ca>, application takes students another step closer to the real business environment. It reinforces and extends learning from RIT simulations by enabling students to apply what they have learned from simulation to real-time quotes for actual market securities. This requires students to learn important institutional details, such as managing currency choice, contract sizes, financing and margin requirements, transactions costs, etc. The RPM enables students to implement, track and manage a portfolio of financial securities using real-time data feeds, to experiment with deriving asset-allocation decisions governed by various objectives, and to evaluate and attribute performance outcomes. The RPM application allows students to focus on the portfolio allocation and risk management decisions by automating the back-office processes of portfolio management.

Figure 5 illustrates an example screen shot from RPM showing an attribution analysis, by sector, of a student's portfolio returns. Among other performance metrics, they can compare their outcomes to an instructor defined benchmark, the class average, or even

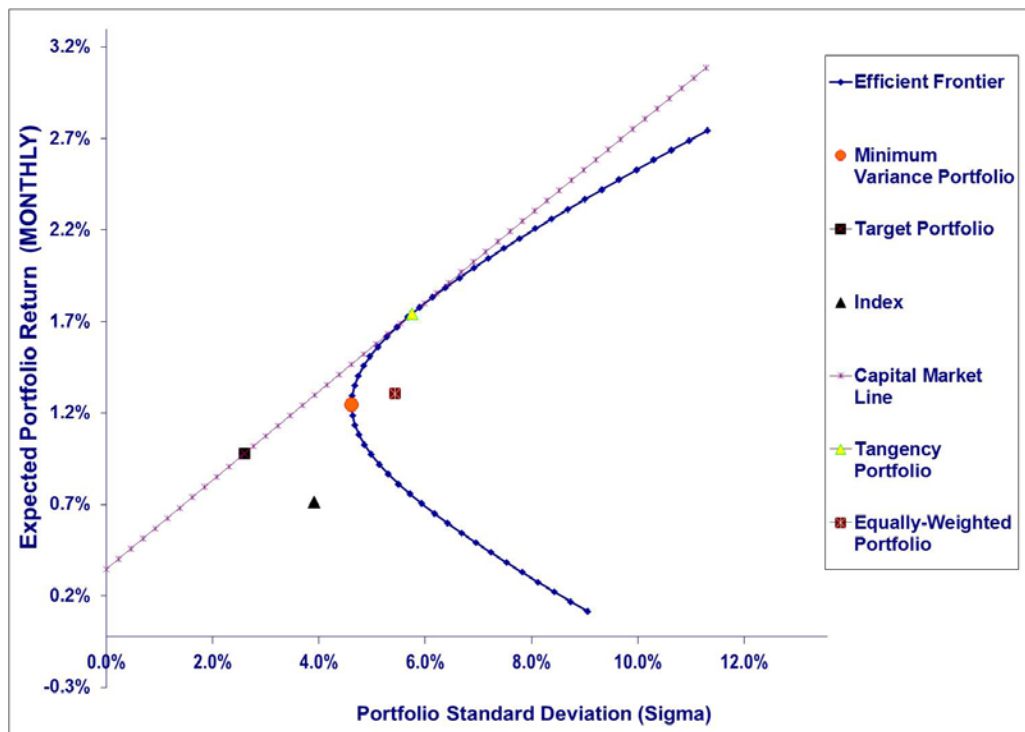
Figure 5: Attribution of Portfolio Returns by Sector



other individuals or groups in the class. These comparisons always lead to a competitive spirit and valuable discussions about differences which serve to clarify the theory and their own applications.

The *Rotman Portfolio Chooser* application provides a platform for students to apply classroom theory about portfolio and risk management to historical and/or real-time data of their choice. The simulations illustrate the application of classroom portfolio and risk management theory, such as the benefits of portfolio diversification, computation of optimal portfolio weights for various alternative objectives based on the risk tolerance of the user, computation of potential losses, portfolio Value-at-Risk for capital adequacy, etc. Students expand their experience by practicing with a range of scenarios and the benefit of immediate feedback on their actions. Figure 5 illustrates the tradeoff between risk and expected return for alternative choices of portfolio weights. In this case, the tangency portfolio maximizes the expected return per unit risk (Sharpe Ratio). In the Portfolio Chooser application, this Excel sheet is then linked to a Monte Carlo simulation of the potential distribution of profits and losses so that the decision maker can evaluate the portfolio Value-at-Risk and associated capital adequacy for risk management.

Figure 5: Risk and Expected Return for a User-Defined Portfolio relative to Benchmarks



SOME EDUCATIONAL BENEFITS OF INTERACTIVE SIMULATIONS

As the FRTL example illustrates, interactive simulations can be designed to complement classroom teaching in a variety of ways. In the following sections, we will describe how educational principles highlighted by research on student learning can be embedded in interactive simulations to strengthen and extend student learning in areas that are critical for innovation (addressing the calls from AACSB and by Rousseau and McCarthy, 2007).

Facilitates Deep Learning

Differences in performance across students are often attributed to differences in individual characteristics such as motivation or ability. It is less commonly appreciated that students hold varying conceptions of knowledge and how it is acquired, and that these conceptions can influence how students learn and whether their learning is effective. Marton and Säljö (1976) derived the distinction between what they called ‘deep’ and ‘surface’ approaches to learning from research with university students. Development of empirical measures and experimental investigations of approaches to learning have shown that students’ emphasis on deep versus surface approaches (in which students emphasize learning for meaning versus memorization, respectively) can be adapted in response to a variety of conditions which are under the control of instructors. For example, course-related variables such as excessive workload, ‘busy work’, and assessments which emphasize recall of superficial information, have all been associated with increases in measures of surface approaches to learning which are accompanied by poor comprehension and recall of course material.

Conversely, students benefit from teaching methods which direct their attention and learning strategies to the deep cognitive processes required for long-lasting learning.³ Well-designed simulations provide environments which are well-suited for such learning. Voogt and van den Akker (2002) comment that simulations can ‘...serve as a bridge between reality and the student’s

³ Note that ‘active’ learning is not necessarily conducive to deep learning. Students can be actively engaged in superficial tasks with little if any educational benefit!

mental model of reality’: simulations can be used to demonstrate characteristic features of models and how relationships between variables operate, thereby helping students develop deep, conceptual understanding of abstract theories and models. Simulations also help to reinforce and extend learning beyond solving textbook problems by enabling students to test their initial understanding of models, try out alternative ideas and observe their consequences. Simulations complement conventional teaching and assessment methods by enabling students to develop the operational knowledge (cognitive strategies and intellectual skills) needed to apply models to novel situations, make decisions that accomplish particular management goals, and develop the capabilities required for life-long learning (self-regulatory processes).

Clarifies Troublesome and Threshold Concepts

Simulations can be especially helpful for teaching a subset of business concepts which are especially difficult for students to grasp. Meyer and Land (2005) noted that some concepts are intrinsically more difficult to learn than others, and that such difficult (or ‘troublesome’) concepts can be identified through recurrent patterns of errors and misconceptions in student work.

Troublesome concepts typically include complex abstract concepts, counterintuitive ideas and interactions between two or more variables (as in many management models). Examples of troublesome concepts have been reported for marketing (Lye, 2006), and for quantitative finance, for example, ‘change of measure’ (Diamond and Smith, 2011).

Meyer and Land also described ‘threshold concepts’ (which, when understood, result in a significant or ‘transformational’ change in understanding or perspective on an issue) and ‘delayed understanding’ (complex concepts which must be built up over time and cannot be comprehended immediately). For example, using RIT simulation cases to learn how to manage liquidity risk provides a very effective way for students to grasp the threshold concept of a ‘marketable limit order’ which is an essential strategy for minimizing price impact (a cost of illiquidity in the market). Identification of such concepts enables instructors to focus students’ time and attention where the

payoff – in terms of understanding of difficult but important concepts -- is likely to be greatest.

Simulations provide an especially powerful way to direct and facilitate learning of these concepts.

Provides Motivation and Opportunities for Practice

Educational researchers have become interested in simulations of real world tasks for learning precisely because the ability to use knowledge in the real world is so important in the knowledge economy. Lombardi (2007) highlighted educational benefits including improvements in student engagement, increased motivation to persist with practice to achieve high levels of performance, integration of different kinds of learning and skills, improvements in communication and collaborative skills, and increased ability to recognize and solve novel problems. It is difficult to find empirical and comparative research on the impact of simulations in the management education literature; however, a recent systematic review of simulation research in health professional education reported consistent gains in students' performance and confidence associated with the use of simulations for learning (Harder, 2010).

Learning how to apply theoretical knowledge to management tasks and scenarios can be challenging. Students must clarify their understanding, work out the connections between their theoretical understanding and relevant aspects of the simulated environment, and determine how these connections inform actions. Interactive simulations can provide several types of support to help students meet the cognitive demands of applying new knowledge.

As in non-stochastic models, simulations can abstract from unnecessary details in order to focus students' attention on those components most important for the learning objective. However, unlike deterministic or non-stochastic models, simulations involving multiple replications drawn from the range of potential scenarios, not only leads to a better appreciation of the implications of uncertainty but also allows students to practice deriving strategies that work well systematically, that is, across the whole range of potential outcomes.

As Vermundt (2003) pointed out, students must be given early and frequent opportunities to

connect theory with practical experience so that the two kinds of knowledge can be integrated into a coherent knowledge base. Performing a new task requires students to process a lot of information – they must keep the theory in mind and make connections between it and the simulation – i.e. new tasks carry a high intrinsic cognitive load. Practice helps students to become familiar with the material and so reduces the intrinsic cognitive load, thereby freeing up cognitive resources for deep learning. Increasing the relative proportion of cognitive resources allocated to deep learning can result in students learning more quickly and/or more effectively. Simulations also help students learn to apply theory by providing opportunities for both guided and independent practice of new skills.

Allows Immediate Feedback and Assessment

Simulations also contribute to gains in learning by providing instant and ongoing formative feedback for students. The findings from meta-analyses of research on feedback (for example, Hattie and Timperley, 2007) report that, in appropriate conditions, the gains in learning associated with formative feedback are substantially larger than the gains associated with many other interventions and variables related to learning.

Of course management education already incorporates extensive faculty and peer feedback. However, simulations provide a complementary source of rich feedback without adding to already heavy faculty workloads: feedback is intrinsic in the way that simulation data, models and systems respond to students' actions when they work through practical examples, make predictions and test hypotheses, or participate in simulations.

Computer-based simulations can be embedded throughout courses to augment conventional teaching and assessment. It is easy to vary task parameters for assessment purposes, and to track and report students' performance. The flexibility adds rigor, reliability and efficiency to assessments. Recent research on assessment indicates that these advantages can in turn be used to improve learning by increasing the frequency of assessments, which encourages students to spend

more time ‘on task’ and to distribute it evenly across the term (Gibbs and Simpson, 2004).

Monitoring students’ performance more closely also allows instructors to respond quickly when students run into difficulty.

Interactive simulations provide an ideal environment for students to learn how to use feedback to drive further learning. Encouraging students to take responsibility for assessing their work and using this information to direct their future learning is important for several reasons. The most immediate reason is that the skills of self-assessment and independent, self-regulated learning are an important component of learning-how-to-learn and will contribute to on-the-job (lifetime) learning. For example, at the completion of each replication of alternative scenarios, the RIT platform has the facility to provide individualized reports to both students and the instructor. In a complex decision-making environment, immediate feedback concerning the outcome of decisions, and how they compare to other participants and to the designated benchmark, provides critical information which students can monitor and use to improve their performance (that is, to develop the capability for self-regulated independent learning).

As future professionals, management students also need to learn how to evaluate complex work in order to assess projects and the contributions of team members. To do so, they must develop an understanding of the characteristics of high quality work, and learn high level analytical and communication capabilities required to give and to act on feedback. Sadler (2010) recommends that students practice evaluating their own and other’s performance in progressively complex tasks. Simulations facilitate students’ self- and peer-evaluations because they provide continuous information about the status of complex environments and tasks. The availability of this information allows students to assess the consequences of their actions and to take appropriate action in response, and to give strategic feedback to peers engaged in collaborative tasks. The information-rich environment inherently supports students in learning to become actively involved in seeking and giving feedback, as recommended by Nicol (2009).

Further, feedback in simulations can be seen as both a stimulus for, and as part of, the process of learning. In their call for a culture of evidence-based management, Rousseau and McCarthy (2007) relate the need for management education to inculcate the use of evidence for self-assessment and for finding ways to improve performance. Interactive simulations provide powerful learning environments for developing the prerequisite self-regulatory skills and attitudes: this in turn would promote the goals of innovation stressed by the AACSB.

CONCLUDING COMMENTS

This article responds to recent calls for evidence-based innovation in management teaching and for business schools to increase their students' capabilities for promoting and managing innovation. Students need to go beyond learning to understand material conceptually; they need to learn how to use their knowledge and skills in complex, real-time environments with uncertainty. Since the management problems associated with innovations are dynamic and often novel, students need to learn 'how to learn' and to make high level evaluation of their own and others' performance. This requires learning regulatory processes to direct and control one's own learning (for example, monitoring one's own behavior, tracking progress, and advance planning).

The AACSB Task Report (2010) referred to the need for business education to adopt pedagogical methods better suited to help students develop the skills to use what they have learned when they enter the workplace. Similarly, Rousseau and McCarthy (2007) advocate the necessity of practice for specific competencies (such as complex problem solving), integration of skills across different domains, and, for teaching students how to learn from experience. Interactive simulations that require students to apply knowledge in order to achieve specific management objectives are especially useful in this respect. Simulations can extend students' learning about theory by providing experience of ill-defined problems in uncertain, dynamic environments, which resemble work in real world business environments. Simulations require students to integrate skills appropriately across the variety of tasks associated with managing problems over time, such as

analyzing complex situations and alternatives, finding information, problem solving, and applying or adapting relevant methods and tools.

We describe an innovative simulation lab integrated into business curricula as an example of a powerful learning environment, and show how interactive simulations integrate core findings from research on learning to improve the quality and kinds of learning that can be achieved. The interactive nature of the custom designed simulations we used as examples, allow participants to receive immediate feedback on the outcomes of their decisions and thus learn from their own level. In addition, students' decisions interact with those of other participants in the simulation so they learn from the actions of others and also experience the behavioral aspects of decision making in an uncertain environment.

The advent of simulation platforms and tools enable faculty to integrate simulation with other teaching methods across the curriculum with the added benefit that the educational benefits are reinforced when simulations are used on a recurrent basis in courses, extra-curricular events and/or independently accessed by students. We argue that simulations can make an important contribution to the development of the new skills demanded of contemporary management graduates without incurring significant costs in terms of faculty workload or disruption of existing curricula. Students will be better prepared for careers and may also become contributors to the creative process that underlies progress in diverse areas of human endeavor. In business education, as in business, the knowledge and skills to adapt to new situations and to implement innovation effectively can be as important as having innovative and creative ideas and aspirations.

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